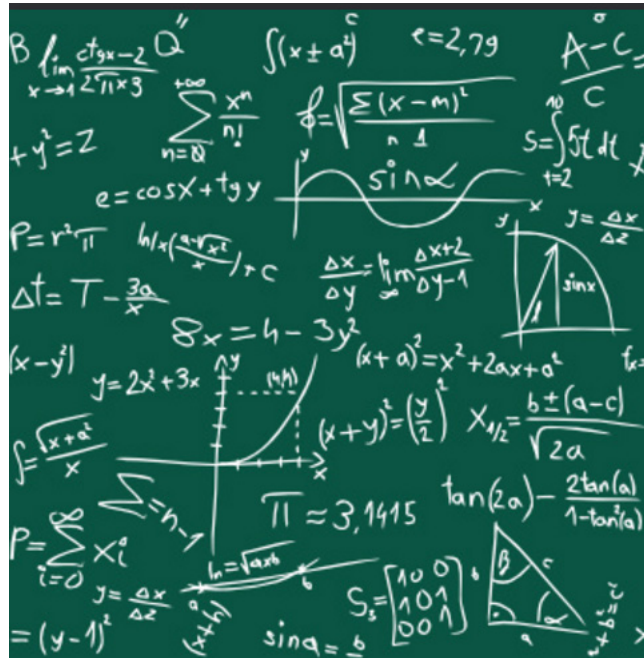


Exercises: computing limits of sequences



- **Authors:** Fabio Furini and Gianluca Priori.
- **Book:** “Esercitazioni Analisi matematica 1”, Marco Bramanti, Esculapio.

Contents

1	Computing limits with basic techniques	2
2	Computing limits using the hierarchy of infinities	7
3	Computing limits using asymptotic estimates	10
4	Computing limits with the sequence converging to Euler's number e	13
5	Computing limits with the comparison theorem	17
6	Computing limits with the ratio test	18
7	Discussion depending on a parameter	21
8	Multiple choice exercises	23

1 Computing limits with basic techniques

If:

$$a_n \rightarrow \ell_a \in \mathbb{R} \quad \text{and} \quad b_n \rightarrow \ell_b \in \mathbb{R},$$

we have:

$$a_n \pm b_n \rightarrow \ell_a \pm \ell_b, \quad \frac{a_n}{b_n} \rightarrow \frac{\ell_a}{\ell_b} \quad (b_n, \ell_b \neq 0, \text{ eventually}),$$

$$a_n b_n \rightarrow \ell_a \ell_b, \quad a_n^{b_n} \rightarrow \ell_a^{\ell_b} \quad (a_n, \ell_a > 0, \text{ eventually}).$$

If:

$$a_n \rightarrow \ell_a \in \mathbb{R}, \quad b_n \rightarrow +\infty \quad \text{and} \quad c_n \rightarrow +\infty,$$

we have:

$$a_n + b_n \rightarrow \ell_a + \infty = +\infty, \quad a_n - b_n \rightarrow \ell_a - \infty = -\infty,$$

$$b_n + c_n \rightarrow +\infty + \infty = +\infty, \quad -b_n - c_n \rightarrow -\infty - \infty = -\infty.$$

If:

$$a_n \rightarrow \ell_a \in \mathbb{R}, \quad b_n \rightarrow 0 \quad \text{and} \quad c_n \rightarrow \infty,$$

we have:

$$a_n c_n \rightarrow \ell_a \infty = \infty \quad (\ell_a \neq 0), \quad \frac{a_n}{b_n} \rightarrow \frac{\ell_a}{0} = \infty \quad (\ell_a \neq 0), \quad \frac{a_n}{c_n} \rightarrow \frac{\ell_a}{\infty} = 0.$$

The four main indeterminate forms are:

$$[+\infty - \infty], \quad [0 \cdot \infty], \quad \left[\frac{0}{0} \right] \quad \text{and} \quad \left[\frac{\infty}{\infty} \right].$$

There are also three other indeterminate forms derived from $[0 \cdot \infty]$:

$$[1^\infty], \quad [0^0] \quad \text{and} \quad [\infty^0].$$

We have:

$$\lim_{n \rightarrow +\infty} n^\alpha = \begin{cases} +\infty & \text{if } \alpha > 0 \\ 1 & \text{if } \alpha = 0 \\ 0 & \text{if } \alpha < 0 \end{cases} \quad \lim_{n \rightarrow +\infty} a^n = \begin{cases} +\infty & \text{if } a > 1 \\ 1 & \text{if } a = 1 \\ 0 & \text{if } |a| < 1 \\ \text{does not exist} & \text{if } a \leq -1 \end{cases}$$

Exercise 1

Compute the following limit, which is an (elementary) indeterminate form:

$$\lim_{n \rightarrow +\infty} \frac{n^2 + 2n}{n + 1} = \left[\frac{\infty}{\infty} \right]$$

$$\lim_{n \rightarrow +\infty} \frac{n^2 + 2n}{n + 1} = \lim_{n \rightarrow +\infty} \frac{\overbrace{n^2}^{\rightarrow 1} \left(1 + \overbrace{\frac{2}{n}}^{\rightarrow 0} \right)}{\underbrace{n}_{\rightarrow 1} \left(1 + \underbrace{\frac{1}{n}}_{\rightarrow 0} \right)} = \lim_{n \rightarrow +\infty} \frac{n^2}{n} = \lim_{n \rightarrow +\infty} n = +\infty.$$

Exercise 2

Compute the following limit, which is an (elementary) indeterminate form:

$$\lim_{n \rightarrow +\infty} \frac{n^4 + 5}{n^5 + 7n - 1} = \left[\frac{\infty}{\infty} \right]$$

$$\lim_{n \rightarrow +\infty} \frac{n^4 + 5}{n^5 + 7n - 1} = \lim_{n \rightarrow +\infty} \frac{n^4 \left(1 + \frac{5}{n^4} \right)}{n^5 \left(1 + \frac{7}{n^4} - \frac{1}{n^5} \right)} = \lim_{n \rightarrow +\infty} \frac{n^4}{n^5} = \lim_{n \rightarrow +\infty} \frac{1}{n} = 0.$$

Exercise 3

Compute the following limit, which is an (elementary) indeterminate form:

$$\lim_{n \rightarrow +\infty} \frac{1 - n^2}{(n + 2)^2} = \left[\frac{\infty}{\infty} \right]$$

$$\lim_{n \rightarrow +\infty} \frac{1 - n^2}{(n + 2)^2} = \lim_{n \rightarrow +\infty} \frac{-n^2 \left(1 - \frac{1}{n^2} \right)}{n^2 \left(1 + \frac{4}{n} + \frac{4}{n^2} \right)} = \lim_{n \rightarrow +\infty} \frac{-n^2}{n^2} = \lim_{n \rightarrow +\infty} -1 = -1.$$

Exercise 4

Compute the following limit, which is an (elementary) indeterminate form:

$$\lim_{n \rightarrow +\infty} \sqrt{n^2 + 1} - \sqrt{n} = [+ \infty - \infty]$$

$$\lim_{n \rightarrow +\infty} \sqrt{n^2 + 1} - \sqrt{n} = \lim_{n \rightarrow +\infty} n \left(\underbrace{\sqrt{1 + \underbrace{\frac{1}{n^2}}_{\rightarrow 0}}}_{\rightarrow 1} - \underbrace{\sqrt{\frac{1}{n}}}_{\rightarrow 0} \right) = \lim_{n \rightarrow +\infty} n = +\infty.$$

Alternative method: multiplying and dividing by $\sqrt{n^2+1} + \sqrt{n}$ we obtain

$$\frac{(\sqrt{n^2+1}-\sqrt{n})(\sqrt{n^2+1}+\sqrt{n})}{\sqrt{n^2+1}+\sqrt{n}} = \frac{(\sqrt{n^2+1})^2 - (\sqrt{n})^2}{\sqrt{n^2+1}+\sqrt{n}} = \frac{n^2+1-n}{\sqrt{n^2+1}+\sqrt{n}},$$

recalling that $(a-b)(a+b) = a^2 - b^2$. Moreover, we have

$$\frac{n^2+1-n}{\sqrt{n^2+1}+\sqrt{n}} = \frac{n^2\left(1+\frac{1}{n^2}-\frac{1}{n}\right)}{\sqrt{n^2\left(1+\frac{1}{n^2}\right)}+\sqrt{n}} = \frac{n^2\left(1+\frac{1}{n^2}-\frac{1}{n}\right)}{n\sqrt{\left(1+\frac{1}{n^2}\right)}+n^{\frac{1}{2}}} = \frac{n^2\left(1+\frac{1}{n^2}-\frac{1}{n}\right)}{n\left(\sqrt{\left(1+\frac{1}{n^2}\right)}+\frac{1}{n^{1/2}}\right)}.$$

Hence

$$\lim_{n \rightarrow +\infty} \sqrt{n^2+1} - \sqrt{n} = \lim_{n \rightarrow +\infty} n \frac{\overbrace{\left(1+\frac{1}{n^2}-\frac{1}{n}\right)}^{-1}}{\underbrace{\left(\sqrt{\left(1+\frac{1}{n^2}\right)}+\frac{1}{n^{1/2}}\right)}_{\rightarrow 1}} = \lim_{n \rightarrow +\infty} n = +\infty.$$

Exercise 5

Compute the following limit, which is an (elementary) indeterminate form:

$$\lim_{n \rightarrow +\infty} e^n - 2^n = [+ \infty - \infty]$$

$$\lim_{n \rightarrow +\infty} e^n - 2^n = \lim_{n \rightarrow +\infty} e^n \underbrace{\left(1 - \underbrace{\left(\frac{2}{e}\right)^n}_{\rightarrow 0}\right)}_{\rightarrow 1} = \lim_{n \rightarrow +\infty} e^n = +\infty$$

since

$$e^n \rightarrow +\infty \text{ and } \left(\frac{2}{e}\right)^n \rightarrow 0.$$

Exercise 6

Compute the following limit, which is an (elementary) indeterminate form:

$$\lim_{n \rightarrow +\infty} 3^n + 4^n - 5^n = [+ \infty + \infty - \infty] = [+ \infty - \infty]$$

$$\lim_{n \rightarrow +\infty} 3^n + 4^n - 5^n = \lim_{n \rightarrow +\infty} -5^n \underbrace{\left(1 - \underbrace{\left(\frac{3}{5} \right)^n}_{\rightarrow 0} - \underbrace{\left(\frac{4}{5} \right)^n}_{\rightarrow 0} \right)}_{\rightarrow 1} = \lim_{n \rightarrow +\infty} -5^n = -\infty$$

since

$$-5^n \rightarrow -\infty, \quad \left(\frac{3}{5} \right)^n \rightarrow 0 \text{ and } \left(\frac{4}{5} \right)^n \rightarrow 0.$$

Exercise 7

Compute the following simple limit

$$\lim_{n \rightarrow +\infty} n^{\sqrt{2}}$$

For every $\alpha > 0$, even irrational, $\lim_{n \rightarrow +\infty} n^\alpha = +\infty$.

Exercise 8

Compute the following simple limit

$$\lim_{n \rightarrow +\infty} n^{-e}$$

$$\lim_{n \rightarrow +\infty} n^{-e} = \lim_{n \rightarrow +\infty} \frac{1}{n^e} = 0.$$

Exercise 9

Compute the following simple limit

$$\lim_{n \rightarrow +\infty} \left(\frac{3n+5}{n^2+120} \right)^{2n}$$

We have

$$\frac{3n+5}{n^2+120} = \frac{3n\left(1+\frac{5}{3n}\right)}{n^2\left(1+\frac{120}{n^2}\right)} = \frac{3}{n} \frac{\left(1+\frac{5}{3n}\right)}{\left(1+\frac{120}{n^2}\right)} \rightarrow 0 \text{ and } 2n \rightarrow +\infty.$$

Hence

$$\lim_{n \rightarrow +\infty} \left(\frac{3n+5}{n^2+120} \right)^{2n} = 0^{+\infty} = 0.$$

Exercise 10

Compute the following simple limit

$$\lim_{n \rightarrow +\infty} \left(\frac{10n^2 - 5n}{n^3 - 8} \right)^{-3n+8}$$

We have

$$\frac{10n^2 - 5n}{n^3 - 8} = \frac{10n^2\left(1 - \frac{5n}{10n^2}\right)}{n^3\left(1 - \frac{8}{n^3}\right)} = \frac{10}{n} \frac{\left(1 - \frac{5}{10n}\right)}{\left(1 - \frac{8}{n^3}\right)} \rightarrow 0 \text{ and } -3n+8 \rightarrow -\infty.$$

Hence

$$\lim_{n \rightarrow +\infty} \left(\frac{10n^2 - 5n}{n^3 - 8} \right)^{-3n+8} = 0^{-\infty} = \frac{1}{0^{+\infty}} = \frac{1}{0} = +\infty.$$

2 Computing limits using the hierarchy of infinities

For every $a > 1$ and $\alpha > 0$ we have:

$$\lim_{n \rightarrow +\infty} \frac{\log_a n}{n^\alpha} = 0, \quad \lim_{n \rightarrow +\infty} \frac{n^\alpha}{a^n} = 0,$$

$$\lim_{n \rightarrow +\infty} \frac{a^n}{n!} = 0, \quad \lim_{n \rightarrow +\infty} \frac{n!}{n^n} = 0.$$

Hence the following infinities are listed in increasing order:

$$\log n, \quad n^\alpha, \quad a^n, \quad n!, \quad n^n.$$

Exercise 11

Compute the limit of the following sequence, if it exists

$$a_n = \frac{2^{1/n} + n^2 + 3^{-n}}{\log^6 n + 2 + n}$$

Consider the numerator; we have:

$$2^{1/n} \rightarrow 1, \quad n^2 \rightarrow +\infty, \quad 3^{-n} \rightarrow 0$$

The power of n with the highest degree is n^2 , and moreover we have

$$\frac{3^{-n}}{n^2} = \frac{1}{3^n n^2} \quad \text{hence} \quad \frac{3^{-n}}{n^2} \rightarrow 0 \quad \text{and} \quad \frac{2^{1/n}}{n^2} \rightarrow 0.$$

Consider the denominator; we have:

$$\log^6 n \rightarrow +\infty, \quad 2 \rightarrow 2, \quad n \rightarrow +\infty$$

By the hierarchy of infinities theorem, we have

$$\frac{\log^6 n}{n} = \left(\underbrace{\frac{\log n}{n^{1/6}}}_{\rightarrow 0} \right)^6 \rightarrow 0.$$

We therefore factor out n^2 in the numerator and n in the denominator and obtain:

$$\begin{aligned} \frac{n^2 \left(1 + \frac{2^{1/n}}{n^2} + \frac{3^{-n}}{n^2} \right)}{n \left(1 + \frac{\log^6 n}{n} + \frac{2}{n} \right)} &= n \frac{\left(1 + \frac{\overbrace{2^{1/n}}^{\rightarrow 0}}{n^2} + \frac{\overbrace{3^{-n}}^{\rightarrow 0}}{n^2} \right)}{\left(1 + \underbrace{\frac{\log^6 n}{n}}_{\rightarrow 0} + \underbrace{\frac{2}{n}}_{\rightarrow 0} \right)} \\ \lim_{n \rightarrow +\infty} \frac{2^{1/n} + n^2 + 3^{-n}}{\log^6 n + 2 + n} &= \lim_{n \rightarrow +\infty} n \frac{\overbrace{\left(1 + \frac{2^{1/n}}{n^2} + \frac{3^{-n}}{n^2} \right)}^{\rightarrow 1}}{\left(1 + \frac{\log^6 n}{n} + \frac{2}{n} \right)} = +\infty \end{aligned}$$

Exercise 12

Compute the limit of the following sequence, if it exists

$$a_n = \frac{3n^3 + 3^n + \log n}{\log^6 n + 2^{2n} + n^5}$$

Consider the numerator; we have:

$$3n^3 \rightarrow +\infty, \quad 3^n \rightarrow +\infty, \quad \log n \rightarrow +\infty$$

that is, a sum of sequences tending to infinity. We also have:

$$\frac{3n^3}{3^n} \rightarrow 0 \quad \text{and} \quad \frac{\log n}{3^n} \rightarrow 0,$$

by the hierarchy of infinities theorem. Hence the principal part is 3^n . Consider the denominator; we have:

$$\log^6 n \rightarrow +\infty, \quad 2^{2n} = 4^n \rightarrow +\infty, \quad n^5 \rightarrow +\infty$$

that is, a sum of sequences tending to infinity. We also have:

$$\frac{\log^6 n}{4^n} \rightarrow 0 \quad \text{and} \quad \frac{n^5}{4^n} \rightarrow 0,$$

by the hierarchy of infinities theorem. Hence the principal part is 4^n .

We therefore factor out 3^n in the numerator and 4^n in the denominator and obtain:

$$\frac{3^n \left(1 + \frac{3n^3}{3^n} + \frac{\log n}{3^n} \right)}{4^n \left(1 + \frac{\log^6 n}{4^n} + \frac{n^5}{4^n} \right)} = \left(\frac{3}{4} \right)^n \frac{\left(1 + \overbrace{\frac{3n^3}{3^n}}^{-0} + \overbrace{\frac{\log n}{3^n}}^{-0} \right)}{\left(1 + \underbrace{\frac{\log^6 n}{4^n}}_{\rightarrow 0} + \underbrace{\frac{n^5}{4^n}}_{\rightarrow 0} \right)}$$

hence

$$\lim_{n \rightarrow +\infty} \frac{3n^3 + 3^n + \log n}{\log^6 n + 2^{2n} + n^5} = \lim_{n \rightarrow +\infty} \left(\frac{3}{4} \right)^n \frac{\left(1 + \frac{3n^3}{3^n} + \frac{\log n}{3^n} \right)}{\underbrace{\left(1 + \frac{\log^6 n}{4^n} + \frac{n^5}{4^n} \right)}_{\rightarrow 1}} = 0$$

Exercise 13

Compute the following simple limit

$$\lim_{n \rightarrow +\infty} \sqrt[n]{n^2}$$

We have

$$\sqrt[n]{n} = n^{\frac{1}{n}} = e^{\log n^{1/n}} = e^{\frac{\log n}{n}}$$

and, studying the sequence in the exponent,

$$a_n = \frac{\log n}{n} \quad \text{we have} \quad a_n \rightarrow 0$$

thanks to the hierarchy of infinities theorem. Hence we have:

$$\lim_{n \rightarrow +\infty} \sqrt[n]{n} = \lim_{n \rightarrow +\infty} e^{\frac{\log n}{n}} = 1.$$

Consequently

$$\lim_{n \rightarrow +\infty} \sqrt[n]{n^2} = \lim_{n \rightarrow +\infty} (\sqrt[n]{n})^2 = 1^2 = 1.$$

3 Computing limits using asymptotic estimates

We have $a_n \sim b_n$ if $\frac{a_n}{b_n} \rightarrow 1$ or if $a_n = b_n c_n$ with $c_n \rightarrow 1$.

Moreover, we have $a_n + b_n \sim a_n$ if $\frac{b_n}{a_n} \rightarrow 0$

Exercise 14

Compute the following limit

$$\lim_{n \rightarrow +\infty} \frac{2^n + n^2 + 1}{5^n + 2^n + n}$$

We have

$$2^n + n^2 + 1 \sim 2^n \text{ since } \frac{n^2}{2^n} \rightarrow 0 \text{ and } \frac{1}{2^n} \rightarrow 0$$

and

$$5^n + 2^n + n \sim 5^n \text{ since } \frac{2^n}{5^n} = \left(\frac{2}{5}\right)^n \rightarrow 0 \text{ and } \frac{n}{5^n} \rightarrow 0$$

hence

$$\lim_{n \rightarrow +\infty} \frac{2^n + n^2 + 1}{5^n + 2^n + n} = \lim_{n \rightarrow +\infty} \frac{2^n}{5^n} = \lim_{n \rightarrow +\infty} \left(\frac{2}{5}\right)^n = 0$$

Exercise 15

Compute the following limit

$$\lim_{n \rightarrow +\infty} \frac{n! - 5^n}{7^n}$$

We have

$$n! - 5^n \sim n! \text{ since } \frac{5^n}{n!} \rightarrow 0$$

hence

$$\lim_{n \rightarrow +\infty} \frac{n! - 5^n}{7^n} = \lim_{n \rightarrow +\infty} \frac{n!}{7^n} = +\infty$$

Exercise 16

Compute the following limit

$$\lim_{n \rightarrow +\infty} \frac{2^{3n-1} - n^2}{(2n)! - 5^n}$$

We have

$$2^{3n-1} - n^2 \sim 2^{3n-1} \text{ since } \frac{n^2}{2^{3n-1}} \rightarrow 0$$

and

$$(2n)! - 5^n \sim (2n)! \text{ since } \frac{5^n}{(2n)!} \rightarrow 0$$

hence

$$\lim_{n \rightarrow +\infty} \frac{2^{3n-1} - n^2}{(2n)! - 5^n} = \lim_{n \rightarrow +\infty} \frac{2^{3n-1}}{(2n)!} = 0$$

Exercise 17

Compute the limit of the following sequence, if it exists

$$\lim_{n \rightarrow +\infty} \frac{3^n}{n! n^{1/n}}$$

We have

$$n! n^{1/n} \sim n! \text{ since } n^{1/n} = \sqrt[n]{n} \rightarrow 1$$

because

$$n^{1/n} = e^{\log n^{1/n}} = e^{\frac{\log n}{n}} \text{ and } \frac{\log n}{n} \rightarrow 0.$$

Hence

$$\lim_{n \rightarrow +\infty} \frac{3^n}{n! n^{1/n}} = \lim_{n \rightarrow +\infty} \frac{3^n}{n!} = 0.$$

Exercise 18

Compute the limit of the following sequence, if it exists

$$\lim_{n \rightarrow +\infty} \frac{\log(1 + e^n)}{\sqrt{1 + n^2}}$$

Observing that

$$\log(1 + e^n) \sim \log(e^n) = n \log e = n$$

we have

$$\lim_{n \rightarrow +\infty} \frac{\log(1 + e^n)}{\sqrt{1 + n^2}} = \lim_{n \rightarrow +\infty} \frac{n}{\sqrt{1 + n^2}} = \lim_{n \rightarrow +\infty} \sqrt{\frac{n^2}{1 + n^2}} = 1$$

because

$$\frac{n^2}{1 + n^2} \rightarrow 1.$$

Exercise 19

Compute the limit of the following sequence, if it exists

$$\lim_{n \rightarrow +\infty} \frac{n + 1}{\log_2(3 + n^n)}$$

Observing that

$$\log_2(3 + n^n) \sim \log_2(n^n) = n \log_2 n$$

we have

$$\lim_{n \rightarrow +\infty} \frac{n+1}{\log_2(3 + n^n)} = \lim_{n \rightarrow +\infty} \frac{n+1}{n \log_2 n} = 0$$

because

$$\frac{n+1}{n} \rightarrow 1 \quad \text{and} \quad \frac{1}{\log_2 n} \rightarrow 0.$$

4 Computing limits with the sequence converging to Euler's number e

For indeterminate forms $1^{\pm\infty}$, one can use:

$$\left(1 + \frac{1}{a_n}\right)^{a_n} \rightarrow e \quad \text{as } a_n \rightarrow \pm\infty.$$

Exercise 20

Compute the following limit

$$\lim_{n \rightarrow +\infty} \left(\frac{n-1}{n-3}\right)^n$$

We have

$$\left(\frac{n-1}{n-3}\right)^n = \left(\frac{n-3-1+3}{n-3}\right)^n = \left(1 + \frac{2}{n-3}\right)^n = \left(1 + \frac{1}{\frac{n-3}{2}}\right)^n$$

hence

$$\lim_{n \rightarrow +\infty} \left(\frac{n-1}{n-3}\right)^n = \lim_{n \rightarrow +\infty} \left[\left(1 + \frac{1}{\frac{n-3}{2}}\right)^{\frac{n-3}{2}} \right]^{\frac{2n}{n-3}} = e^2,$$

since

$$\left(1 + \frac{1}{\frac{n-3}{2}}\right)^{\frac{n-3}{2}} \rightarrow e, \quad 2 \left(\frac{n}{n-3}\right) = 2 \left(\frac{n-3+3}{n-3}\right) = 2 \left(1 + \underbrace{\frac{3}{n-3}}_{\rightarrow 0}\right) \rightarrow 2$$

and

$$\frac{n-3}{2} \rightarrow +\infty.$$

Exercise 21

Compute the following limit

$$\lim_{n \rightarrow +\infty} \left(\frac{n^2+1}{n^2}\right)^n$$

$$\lim_{n \rightarrow +\infty} \left(\frac{n^2+1}{n^2}\right)^n = \lim_{n \rightarrow +\infty} \left[\left(1 + \frac{1}{n^2}\right)^{n^2} \right]^{\frac{1}{n}} = e^0 = 1$$

since

$$\left(1 + \frac{1}{n^2}\right)^{n^2} \rightarrow e, \quad \frac{1}{n} \rightarrow 0$$

and

$$n^2 \rightarrow +\infty.$$

Exercise 22

Compute the following limit

$$\lim_{n \rightarrow +\infty} \left(\frac{2n-5}{2n}\right)^{-n}$$

$$\lim_{n \rightarrow +\infty} \left(\frac{2n-5}{2n} \right)^{-n} = \lim_{n \rightarrow +\infty} \left[\left(1 + \frac{1}{-\frac{2n}{5}} \right)^{-\frac{2n}{5}} \right]^{\frac{5}{2}} = \sqrt{e^5}.$$

since

$$\left(1 + \frac{1}{-\frac{2n}{5}} \right)^{-\frac{2n}{5}} \rightarrow e \quad \text{and} \quad -\frac{2n}{5} \rightarrow -\infty.$$

Exercise 23

Compute the following limit

$$\lim_{n \rightarrow +\infty} \frac{n^{n-1}}{(n-1)^n}$$

We have

$$\frac{n^{n-1}}{(n-1)^n} = \frac{1}{n} \left(\frac{n}{n-1} \right)^n = \frac{1}{n} \left(\frac{n-1}{n} \right)^{-n} = \frac{1}{n} \left(1 + \frac{1}{-n} \right)^{-n}$$

hence

$$\lim_{n \rightarrow +\infty} \frac{n^{n-1}}{(n-1)^n} = \lim_{n \rightarrow +\infty} \frac{1}{n} \left(1 + \frac{1}{-n} \right)^{-n} = 0$$

since

$$\left(1 + \frac{1}{-n} \right)^{-n} \rightarrow e \quad \text{and} \quad \frac{1}{n} \rightarrow 0.$$

Exercise 24

Compute the limit of the following sequence, if it exists

$$\lim_{n \rightarrow +\infty} \left(\frac{n^2 + 2}{n^2 + n + 1} \right)^{2n}$$

Since

$$\left(\frac{n^2 + 2}{n^2 + n + 1} \right) \rightarrow 1 \text{ and } 2n \rightarrow +\infty$$

we have an indeterminate form of type $[1^{+\infty}]$.

Since

$$\left(\frac{n^2 + 2}{n^2 + n + 1} \right)^{2n} = \left(\frac{n^2 + n + 1 - n + 1}{n^2 + n + 1} \right)^{2n} = \left(1 + \frac{-n + 1}{n^2 + n + 1} \right)^{2n},$$

we can therefore write

$$\left(1 + \frac{-n + 1}{n^2 + n + 1} \right)^{2n} = \left(1 + \frac{1}{\frac{n^2 + n + 1}{-n + 1}} \right)^{2n} = \left[\left(1 + \frac{1}{\frac{n^2 + n + 1}{-n + 1}} \right)^{\frac{n^2 + n + 1}{-n + 1}} \right]^{\left(\frac{-n + 1}{n^2 + n + 1} \right) 2n}.$$

Hence

$$\lim_{n \rightarrow +\infty} \left(\frac{n^2 + 2}{n^2 + n + 1} \right)^{2n} = \lim_{n \rightarrow +\infty} \left[\left(1 + \frac{1}{\frac{n^2 + n + 1}{-n + 1}} \right)^{\frac{n^2 + n + 1}{-n + 1}} \right]^{\left(\frac{-n + 1}{n^2 + n + 1} \right) 2n} = e^{-2},$$

since

$$\left(1 + \frac{1}{\frac{n^2 + n + 1}{-n + 1}} \right)^{\frac{n^2 + n + 1}{-n + 1}} \rightarrow e, \quad \left(\frac{-n + 1}{n^2 + n + 1} \right) 2n = \left(\frac{\overbrace{-2n^2 + 2n}^{\sim -2n^2}}{\underbrace{n^2 + n + 1}_{\sim n^2}} \right) \rightarrow -2$$

and

$$\frac{n^2 + n + 1}{-n + 1} = \frac{n^2 \overbrace{\left(1 + \frac{1}{n} + \frac{1}{n^2} \right)}^{\rightarrow 1}}{\underbrace{-n \left(1 - \frac{1}{n} \right)}_{\rightarrow -1}} \rightarrow -\infty.$$

5 Computing limits with the comparison theorem

- If $c_n \rightarrow 0$ and $|b_n| \leq c_n$ eventually, then $b_n \rightarrow 0$.
- If $c_n \rightarrow 0$ and b_n is bounded (even if not convergent), then $c_n b_n \rightarrow 0$. The product of an infinitesimal sequence and a bounded one is infinitesimal.
- A sum of infinitesimals and bounded sequences is asymptotically equivalent to the infinity of highest order.

Exercise 25

Compute the following limit

$$\lim_{n \rightarrow +\infty} \frac{n + \sin n}{\cos n + \log n}$$

We have

$$n + \sin n \sim n \text{ since } \frac{\sin n}{n} \rightarrow 0$$

the product of an infinitesimal sequence $\{\frac{1}{n}\}$ and a bounded one $\{\sin n\}$. Moreover,

$$\cos n + \log n \sim \log n \text{ since } \frac{\cos n}{\log n} \rightarrow 0$$

the product of an infinitesimal sequence $\{\frac{1}{\log n}\}$ and a bounded one $\{\cos n\}$. Hence

$$\lim_{n \rightarrow +\infty} \frac{n + \sin n}{\cos n + \log n} = \lim_{n \rightarrow +\infty} \frac{n}{\log n} = +\infty$$

Exercise 26

Compute the limit of the following sequence, if it exists

$$\lim_{n \rightarrow +\infty} \frac{n \sin n + \sin n^2}{n^2 + 1}$$

We have

$$\frac{n \sin n + \sin n^2}{n^2 + 1} = \frac{n \sin n}{n^2 + 1} + \frac{\sin n^2}{n^2 + 1}$$

and

$$\frac{n}{n^2 + 1} \sin n \rightarrow 0 \text{ since } \frac{n}{n^2 + 1} \rightarrow 0 \text{ and } \sin n \text{ is bounded.}$$

Moreover

$$\frac{1}{n^2 + 1} \sin n^2 \rightarrow 0 \text{ since } \frac{1}{n^2 + 1} \rightarrow 0 \text{ and } \sin n^2 \text{ is bounded.}$$

Hence

$$\lim_{n \rightarrow +\infty} \frac{n \sin n + \sin n^2}{n^2 + 1} = \lim_{n \rightarrow +\infty} \underbrace{\frac{n \sin n}{n^2 + 1}}_{\rightarrow 0} + \underbrace{\frac{\sin n^2}{n^2 + 1}}_{\rightarrow 0} = 0$$

Alternative method.

Note that the sequences $\{\sin n\}$ and $\{\sin n^2\}$ are irregular but bounded. Hence we can write the following upper bound

$$\left| \frac{n \sin n + \sin n^2}{n^2 + 1} \right| \leq \frac{n + 1}{n^2 + 1}$$

and we have

$$\lim_{n \rightarrow +\infty} \frac{n + 1}{n^2 + 1} = \lim_{n \rightarrow +\infty} \frac{n}{n^2} = \lim_{n \rightarrow +\infty} \frac{1}{n} = 0.$$

Hence, by the comparison theorem, we have:

$$\lim_{n \rightarrow +\infty} \frac{n \sin n + \sin n^2}{n^2 + 1} = 0$$

6 Computing limits with the ratio test

Given a positive sequence ($a_n > 0$ for every n):

if there exists $\lim_{n \rightarrow +\infty} \frac{a_{n+1}}{a_n} = \ell$ and $\ell < 1$, then $a_n \rightarrow 0$,

if there exists $\lim_{n \rightarrow +\infty} \frac{a_{n+1}}{a_n} = \ell$ and $\ell > 1$ (or $\ell = +\infty$), then $a_n \rightarrow +\infty$.

Exercise 27

Compute the limit of the following sequence, if it exists

$$\lim_{n \rightarrow +\infty} \frac{3^{n-1}}{n(n+1)!}$$

Since this is a sequence with positive terms whose general term is the ratio of simpler sequences of exponential and factorial type, it is natural to apply the ratio test.

We therefore compute:

$$\begin{aligned} \frac{a_{n+1}}{a_n} &= \frac{\frac{3^n}{(n+1) \cdot (n+2)!}}{\frac{3^{n-1}}{n(n+1)!}} = \frac{3^n}{(n+1) \cdot \underbrace{(n+2)!}_{=(n+1)! \cdot (n+2)}} \cdot \frac{n(n+1)!}{3^{n-1}} = \\ &= \frac{3n}{(n+1)(n+2)} = \frac{3n}{n^2 + 2n + n + 2} = \frac{3n}{n^2 \left(1 + \frac{3}{n} + \frac{2}{n^2}\right)} = \frac{3}{n \left(1 + \frac{3}{n} + \frac{2}{n^2}\right)} \end{aligned}$$

We have:

$$\lim_{n \rightarrow +\infty} \frac{a_{n+1}}{a_n} = \lim_{n \rightarrow +\infty} \frac{3}{n \left(1 + \frac{3}{n} + \frac{2}{n^2}\right)} = 0$$

Hence, by the ratio test:

$$\lim_{n \rightarrow +\infty} \frac{3^{n-1}}{n(n+1)!} = 0$$

Exercise 28

Compute the limit of the following sequence, if it exists

$$\lim_{n \rightarrow +\infty} \frac{n^3 2^n}{n!}$$

We have a positive sequence; we compute:

$$\frac{a_{n+1}}{a_n} = \frac{(n+1)^3 2^{(n+1)}}{(n+1)!} \frac{n!}{n^3 2^n} = \underbrace{\left(\frac{n+1}{n}\right)^3}_{\rightarrow 1} \frac{2}{n+1} \sim \frac{2}{n+1}$$

We therefore have:

$$\lim_{n \rightarrow +\infty} \frac{a_{n+1}}{a_n} = \lim_{n \rightarrow +\infty} \frac{2}{n+1} = 0$$

Then, by the ratio test:

$$\lim_{n \rightarrow +\infty} \frac{n^3 2^n}{n!} = 0$$

Exercise 29

Compute the limit of the following sequence, if it exists

$$\lim_{n \rightarrow +\infty} \frac{n^n}{(n+1)!}$$

We have a positive sequence; we compute:

$$\frac{a_{n+1}}{a_n} = \frac{(n+1)^{n+1}}{(n+2)!} \frac{(n+1)!}{n^n} = \underbrace{\frac{n+1}{n+2}}_{\rightarrow 1} \left(\frac{n+1}{n}\right)^n \sim \left(1 + \frac{1}{n}\right)^n$$

We therefore have:

$$\lim_{n \rightarrow +\infty} \frac{a_{n+1}}{a_n} = \lim_{n \rightarrow +\infty} \left(1 + \frac{1}{n}\right)^n = e > 1$$

Then, by the ratio test:

$$\lim_{n \rightarrow +\infty} \frac{n^n}{(n+1)!} = +\infty$$

Exercise 30

Compute the limit of the following sequence, if it exists

$$\lim_{n \rightarrow +\infty} \frac{n^{\frac{1}{n}} 2^n}{(n+1)!}$$

We have:

$$\frac{\overbrace{n^{\frac{1}{n}}}^{\rightarrow 1} 2^n}{(n+1)!} \sim \frac{2^n}{(n+1)!} \equiv b_n$$

We now have the positive sequence $\{b_n\}$; we compute:

$$\frac{b_{n+1}}{b_n} = \frac{2^{n+1}}{(n+2)!} \frac{(n+1)!}{2^n} = \frac{2}{n+2}$$

We therefore have:

$$\lim_{n \rightarrow +\infty} \frac{b_{n+1}}{b_n} = \lim_{n \rightarrow +\infty} \frac{2}{n+2} = 0$$

Then, by the ratio test:

$$\lim_{n \rightarrow +\infty} \frac{n^{\frac{1}{n}} 2^n}{(n+1)!} = 0$$

7 Discussion depending on a parameter

Exercise 31

Compute, as the parameter $\alpha \in \mathbb{R}$ varies, the limit of the sequence

$$\lim_{n \rightarrow +\infty} \frac{5^{3\alpha n}}{2^{3n+1}}$$

We have

$$\frac{5^{3\alpha n}}{2^{3n+1}} = \frac{1}{2} \left(\frac{5^\alpha}{2} \right)^{3n}$$

hence

$$\lim_{n \rightarrow +\infty} \frac{5^{3\alpha n}}{2^{3n+1}} = \lim_{n \rightarrow +\infty} \frac{1}{2} \left(\frac{5^\alpha}{2} \right)^{3n} = \frac{1}{2} \lim_{n \rightarrow +\infty} \left(\frac{5^\alpha}{2} \right)^{3n} = \ell$$

At this point, we can study three cases:

- *Case 1*

$$\frac{5^\alpha}{2} > 1 \iff \alpha > \log_5 2$$

then $\ell = +\infty$;

- *Case 2*

$$0 < \frac{5^\alpha}{2} < 1 \iff \alpha < \log_5 2$$

then $\ell = 0$;

- *Case 3*

$$\frac{5^\alpha}{2} = 1 \iff \alpha = \log_5 2$$

then $\ell = \frac{1}{2}$.

In conclusion:

$$\lim_{n \rightarrow +\infty} \frac{5^{3\alpha n}}{2^{3n+1}} = \begin{cases} +\infty & \text{if } \alpha > \log_5 2 \\ \frac{1}{2} & \text{if } \alpha = \log_5 2 \\ 0 & \text{if } \alpha < \log_5 2 \end{cases}$$

Exercise 32

Compute, as the parameter $\alpha \in \mathbb{R}$ varies, the limit of the sequence

$$\lim_{n \rightarrow +\infty} (e^{(2-\alpha)n} + 1) \log \left(1 + \frac{1}{e^n} \right)$$

We have

$$\begin{aligned} a_n &= (e^{(2-\alpha)n} + 1) \log\left(1 + \frac{1}{e^n}\right) = (e^{(2-\alpha)n} + 1) \frac{e^n}{e^n} \log\left(1 + \frac{1}{e^n}\right) \\ &= (e^{(2-\alpha)n} + 1) \frac{1}{e^n} \log\left(1 + \frac{1}{e^n}\right)^{e^n} \\ &\sim (e^{(2-\alpha)n} + 1) \frac{1}{e^n} \end{aligned}$$

since

$$\left(1 + \frac{1}{e^n}\right)^{e^n} \rightarrow e \quad \text{and} \quad \log\left(1 + \frac{1}{e^n}\right)^{e^n} \rightarrow 1$$

Hence

$$\lim_{n \rightarrow +\infty} a_n = \lim_{n \rightarrow +\infty} (e^{(2-\alpha)n} + 1) \frac{1}{e^n}$$

On the other hand

$$(e^{(2-\alpha)n} + 1) \frac{1}{e^n} = e^{(1-\alpha)n} + e^{-n} \rightarrow \begin{cases} +\infty & \text{if } 1 - \alpha > 0 \\ 1 & \text{if } 1 - \alpha = 0 \\ 0 & \text{if } 1 - \alpha < 0 \end{cases}$$

Summarizing, we obtain

$$\lim_{n \rightarrow +\infty} (e^{(2-\alpha)n} + 1) \log\left(1 + \frac{1}{e^n}\right) = \begin{cases} +\infty & \text{if } \alpha < 1 \\ 1 & \text{if } \alpha = 1 \\ 0 & \text{if } \alpha > 1 \end{cases}$$

8 Multiple choice exercises

Exercise 33

Let

$$\lim_{n \rightarrow +\infty} \left(\frac{\sqrt{3} + n}{e - 2n} \right)^n = \ell$$

- (a) $\ell = 0$
- (b) $\ell = e^{(\sqrt{3} - \frac{e}{2})}$
- (c) $\ell = +\infty$
- (d) None of the other answers is correct.

Setting

$$a_n = \frac{\sqrt{3} + n}{e - 2n}$$

we have $a_n < 0$ for $n \geq 2$ and

$$|a_n|^n = \left(\frac{\sqrt{3} + n}{2n - e} \right)^n \rightarrow 0$$

since $|a_n| \rightarrow 1/2$. From $|a_n|^n \rightarrow 0$ it follows that $(a_n)^n \rightarrow 0$. The correct answer is (a).

Exercise 34

Let

$$\lim_{n \rightarrow +\infty} (-1)^n n^\alpha \left(\frac{1+3n}{2-n} \right)^n = \ell$$

- (a) ℓ does not exist if $\alpha \geq 0$
- (b) $\ell = +\infty$ for every $\alpha \in \mathbb{R}$
- (c) $\ell = 0$ if $\alpha < 0$
- (d) None of the other answers is correct.

We set

$$a_n = (-1)^n \left(\frac{1+3n}{2-n} \right)^n = \left(\frac{1+3n}{n-2} \right)^n.$$

We have $a_n \rightarrow +\infty$ since $(1+3n)/(n-2) \rightarrow 3$. Let us compare a_n with 3^n :

$$\frac{a_n}{3^n} = \left(\frac{1+3n}{3n-6} \right)^n = \left[\left(1 + \frac{1}{\frac{3n-6}{7}} \right)^{\frac{3n-6}{7}} \right]^{\frac{7n}{3n-6}} \rightarrow e^{7/3}$$

from which

$$a_n \sim 3^n e^{7/3}.$$

It follows that

$$\lim_{n \rightarrow +\infty} (-1)^n n^\alpha \left(\frac{1+3n}{2-n} \right)^n = \lim_{n \rightarrow +\infty} e^{7/3} n^\alpha 3^n = +\infty$$

even when $\alpha < 0$, because 3^n is an infinity of higher order than any power of n . The correct answer is (b).

Exercise 35

Let

$$\lim_{n \rightarrow +\infty} \left(\frac{an+2}{en+a} \right)^n = \ell$$

- (a) $\ell = e^{(\frac{2}{e}-1)}$ if $a = e$
- (b) $\ell = 1$ if $a \leq e$
- (c) $\ell = 0$ if $a > 0$
- (d) None of the other answers is correct.

Setting

$$a_n = \frac{an+2}{en+a}$$

we have $a_n \rightarrow a/e$, hence

$$(a_n)^n \rightarrow \begin{cases} +\infty, & a > e \\ 0, & 0 < a < e \end{cases}$$

therefore answers (b) and (c) are not correct. In the case $a = e$ we have an indeterminate form 1^∞ :

$$\left(\frac{en+2}{en+e} \right)^n = \left[\left(1 + \frac{1}{\frac{en+e}{2-e}} \right)^{\frac{en+e}{2-e}} \right]^{\frac{(2-e)n}{en+e}} \rightarrow e^{-1+\frac{2}{e}}.$$

The correct answer is (a).